Treasury Management

Question 1

a)

i) Inventory days = $\frac{\text{Inventory}}{\text{COGS}} \times 365$ = $\frac{3000}{9300} \times 365$ $\frac{1300}{6,600}$

	20x6	20x5

(3) = 118 days 72 days

plus Receivable days:

$\frac{\text{Accounts receivable}}{\text{Sales}} \times 365$ = $\frac{3800}{15600} \times 365$ $\frac{1850}{11,100} \times 365$

(3) = 89 days 61 days

207 133

less:

Payable days:

$\frac{\text{Accounts Payable}}{\text{COGS}}$ = $\frac{2870}{9300} \times 365$ $\frac{1600}{6,600} \times 365$

(3) = 113 days 88 days

+1

b) CCC = 20x6 20x5

207 - 113 = 133 - 88

= 94 days = 45 days

(2)

(2) Explanation

(1)

c) Methods to assess credit standing of customers.

- Bank references
- Trade references
- Published information
- Credit agencies
- Company's own sales records

(2 each method)

d) Annual fee = 0.5% of credit sales

Advance = 80% of receivables

Interest = 8% + 1%

(2)

Admin expenses for 20X6 = 1,000

Administration fee = 0.5% × 3,800

= 190.

(2)

Finance Costs = 9% × 1 × (80% × ~~3 800~~ 15 600)

= 9% × ~~3 640~~ 12,480

= ~~273.60~~ 1 123.

(2)

Total Cost = 190 + ~~273.60~~ 1,123

= ~~463.60~~

= K1 313

(2)

∴ Factoring would be ~~cheaper~~ more expensive as the company would only pay K463.60 compared to K1000 it would spend on admin costs.

(2)

(2)

e)	Invoice Discounting	Factoring
Comparison 2	- facility used to raise working capital finance - Uses outstanding Invoices - More expensive compared to O/D	- facility used to raise working capital finance - uses outstanding Invoices. - More expensive compared to O/D
Contrast. 8	- Non-confidential facility - factors takes over sales ledger - May be without recourse - More expensive compared to Invoice discounting	- confidential facility - financier does not take over sales ledger - always with recourse. - less expensive compared to factoring

Question 2

(a) Annual demand = 4,000 units

Delivery cost per order = K10

Storage cost per unit = K2

$$EOQ = \sqrt{\frac{2aC}{h}}$$

$$= \sqrt{\frac{2 \times 4000 \times 10}{2}}$$ (4)

$$= \sqrt{40000}$$

$$= 200 \text{ units}$$

b) JIT is a system of inventory control where inventory items are purchased just in time to be used in the production process.

- Reason why JIT might be impractical in Zambia:
- Greater distances between the suppliers and the manufacturers.
- State of Infrastructure is poor.
- level of technology not advanced.
- Most raw materials are imported from overseas.

2 each Point

(4)

c) Four Inventory financing methods other than overdrafts:

- Supplier Credit
- Supply chain financing
- Collateralized loans
- Asset Based loans
- Floor planning

2 each point

5

Question 3

Minimum cash balance = K7,500
cost per transaction = K18 (2M)
standard deviation = K1000 per day
Interest = 5.11% / 365
= 0.014% per day

a) $z = 3 \times \left\{ \frac{3}{4} \times \frac{\text{transaction cost} \times \text{variance}}{\text{Interest rate}} \right\}^{1/3}$ (2)

Return point = lower limit + $\frac{\text{spread}}{3}(z)$

$z = 3 \times \left\{ 0.75 \times \frac{18 \times 1000^2}{0.00014} \right\}^{1/3}$ (4)

$= 3 \times \left\{ 0.75 \times \frac{18 \times 1000\,000}{0.00014} \right\}^{1/3}$

$= 3 \times \left\{ 0.75 \times 128,571,428,571.43 \right\}^{1/3}$

$= 3 \times 96,428,571,428.57^{1/3}$

$= 3 \times 4.585$

$= K13,756.98 \approx K13757$

upper limit = 13 757 + 7500 = 21,257 (2)

Return point = $K7500 + \frac{13,757}{3}$

$= K7500 + 4586$ (2)

$= K12\,086$

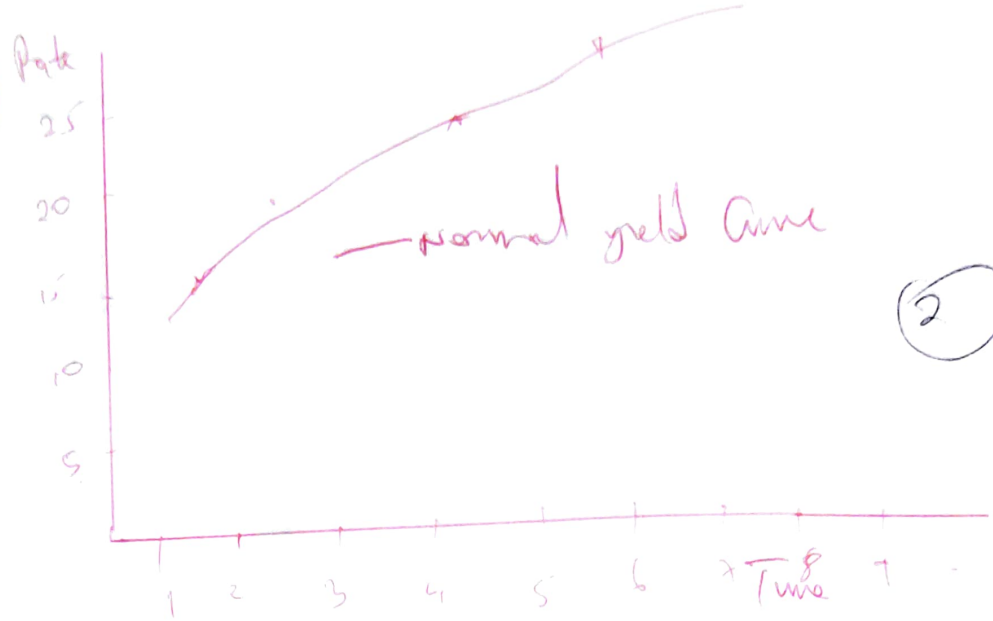
(b) Five investment instruments that ABC Ltd can use to invest surplus cash

- Treasury bills
- Call deposits
- Term deposits
- Certificates of deposit
- Bonds nearing maturity
- Money market accounts

2 each

Question 4

a)



(2)

- normal yield curve
- shows shorter dated tranches yield lower compared to longer dated tranches

(2)

b) - Default risk
- Real capital loss
- Capital loss
- changes in lenders circumstances

(1 each)

b) loan amount = K1 000 000
Interest rate reset = 3 months
Price of interest rate future = 95

i) If interest rate has increased to 8% in September:

Interest rate on loan from bank (8 - 5) = 3% more

(2)

i.e. Interest rate paid to bank =

$1\,000\,000 \times 3\% \times \frac{1}{4} = (K9,500)$

Profit on futures Contract:

XYZ sell contract in June @ 95

Buys contract in Sept @ 92

Profit on futures Contract 3%

(2)

$1\,000\,000 \times 3\% \times \frac{1}{4} = K7,500$

The futures contract has allowed XYZ to offset the increased interest rate it pays to the bank of ~~K3,8~~ K8,300 by making a profit of K8 300 on the futures Contract.

(2)

ii) Features of futures Contract that distinguish it from forwards:

- Traded on an exchange.
- Contracts are standardised in terms of ~~allow~~ ~~to~~ size, settlement etc.
- Requires payment of margin
- No Counterparty risk.

2 each